

Hadoop 2.8.0 installation on Windows 10

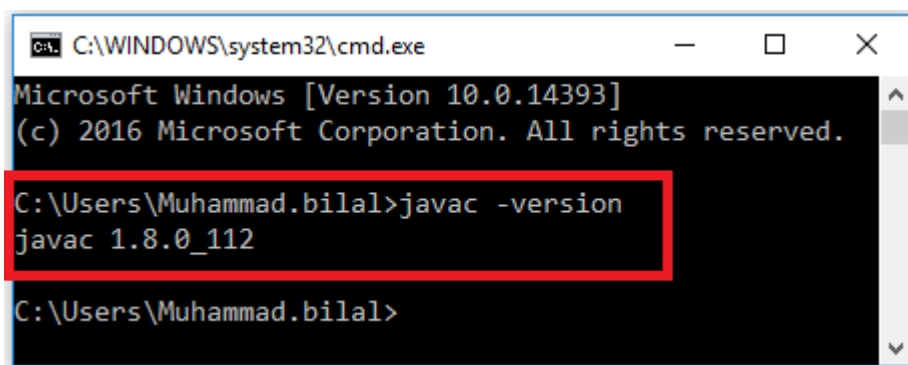
Prepare:

These softwares should be prepared to install Hadoop 2.8.0 on windows 10 64bit

1. Download Hadoop 2.8.0 (Link: <http://www-eu.apache.org/dist/hadoop/common/hadoop-2.8.0/hadoop-2.8.0.tar.gz> OR <http://archive.apache.org/dist/hadoop/core/hadoop-2.8.0/hadoop-2.8.0.tar.gz>)
2. Java JDK 1.8.0.zip (Link: <http://www.oracle.com/technetwork/java/javase/downloads/jdk8-downloads-2133151.html>)

Set up

1. Check either Java 1.8.0 is already installed on your system or not, use "**Javac -version**" to check.

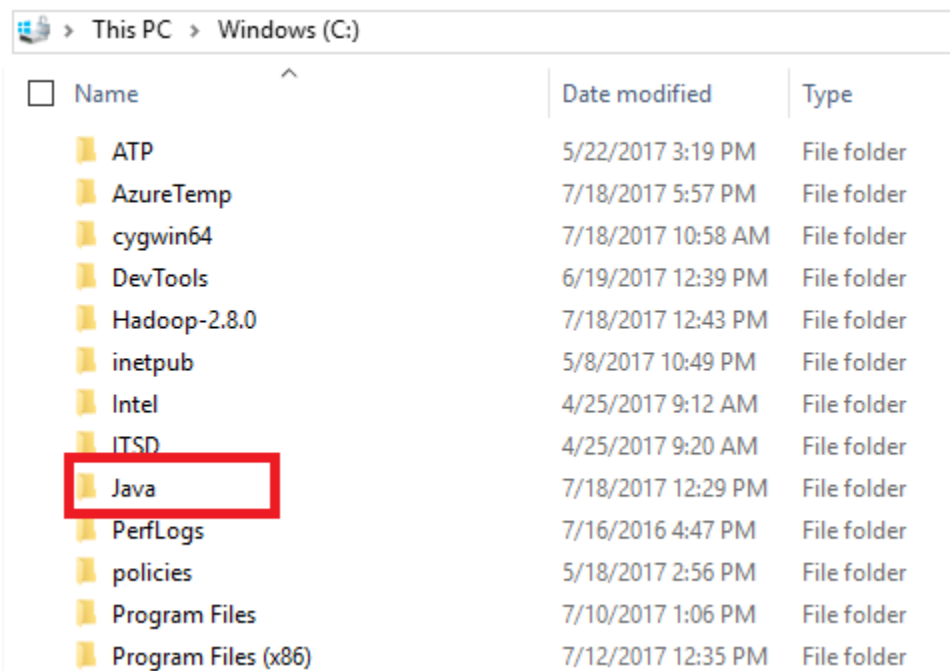


```
C:\WINDOWS\system32\cmd.exe
Microsoft Windows [Version 10.0.14393]
(c) 2016 Microsoft Corporation. All rights reserved.

C:\Users\Muhammad.bilal>javac -version
javac 1.8.0_112

C:\Users\Muhammad.bilal>
```

2. If Java is not installed on your system then first install java

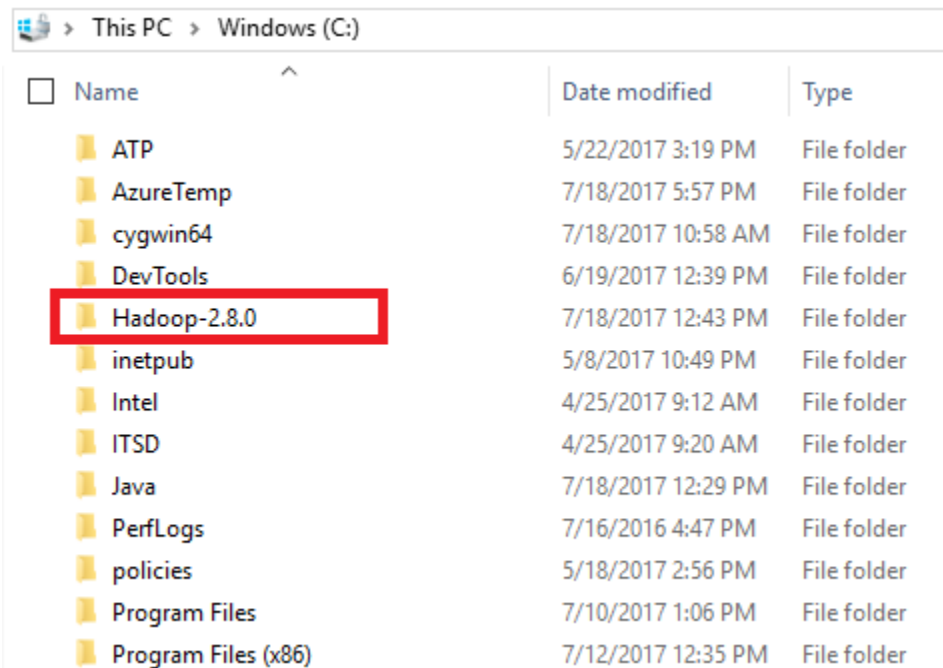


This screenshot shows the Windows File Explorer window for the 'C:\Windows' directory. The 'Java' folder is highlighted with a red rectangle. The table below represents the data shown in the screenshot.

Name	Date modified	Type
ATP	5/22/2017 3:19 PM	File folder
AzureTemp	7/18/2017 5:57 PM	File folder
cygwin64	7/18/2017 10:58 AM	File folder
DevTools	6/19/2017 12:39 PM	File folder
Hadoop-2.8.0	7/18/2017 12:43 PM	File folder
inetpub	5/8/2017 10:49 PM	File folder
Intel	4/25/2017 9:12 AM	File folder
ITSD	4/25/2017 9:20 AM	File folder
Java	7/18/2017 12:29 PM	File folder
PerfLogs	7/16/2016 4:47 PM	File folder
policies	5/18/2017 2:56 PM	File folder
Program Files	7/10/2017 1:06 PM	File folder
Program Files (x86)	7/12/2017 12:35 PM	File folder

under "**C:\JAVA**"

3. Extract file Hadoop 2.8.0.tar.gz or Hadoop-2.8.0.zip and place under "**C:\Hadoop-**

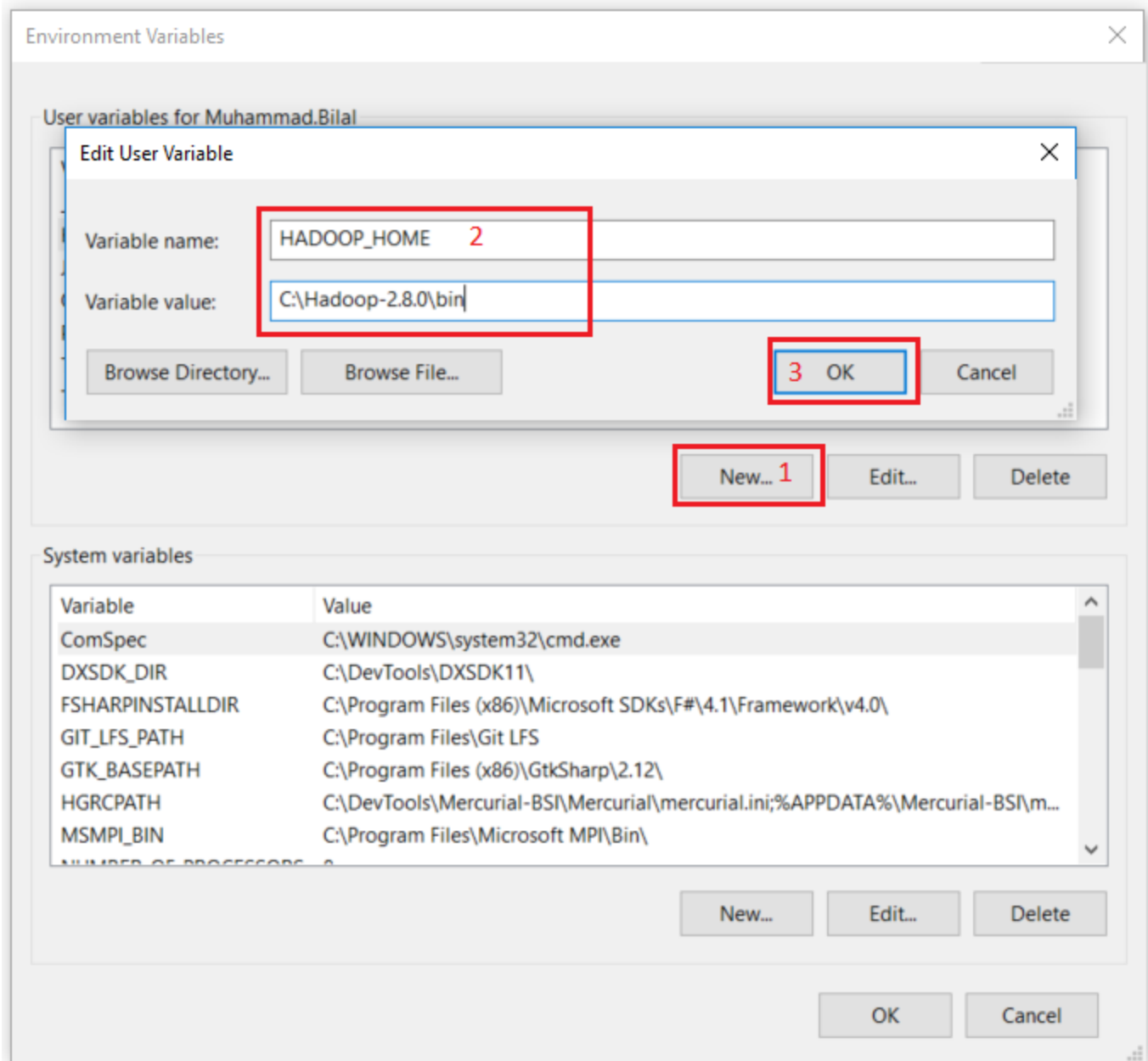


This screenshot shows the Windows File Explorer window for the 'C:\Windows' directory. The 'Hadoop-2.8.0' folder is highlighted with a red rectangle. The table below represents the data shown in the screenshot.

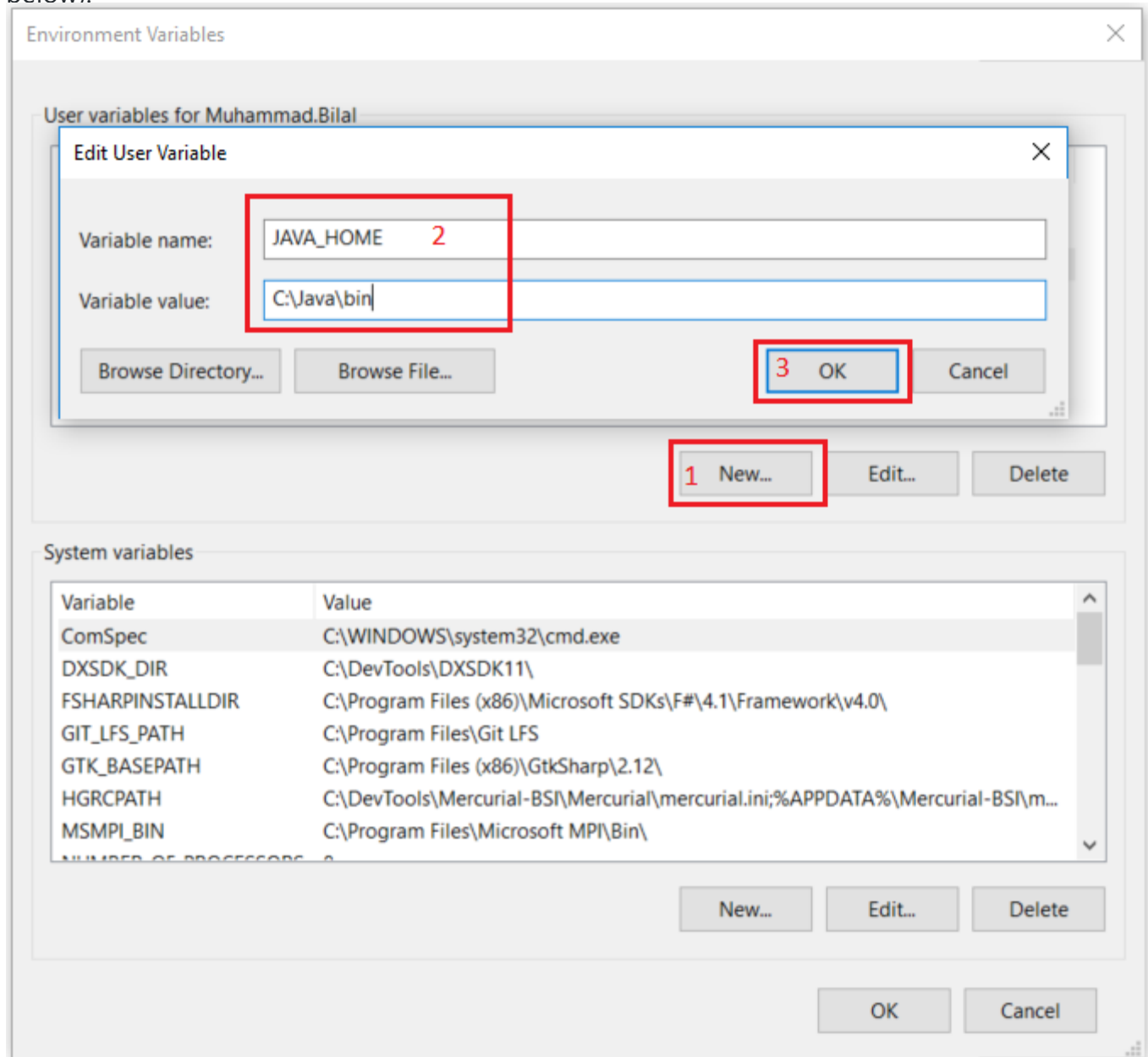
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2.8.0".

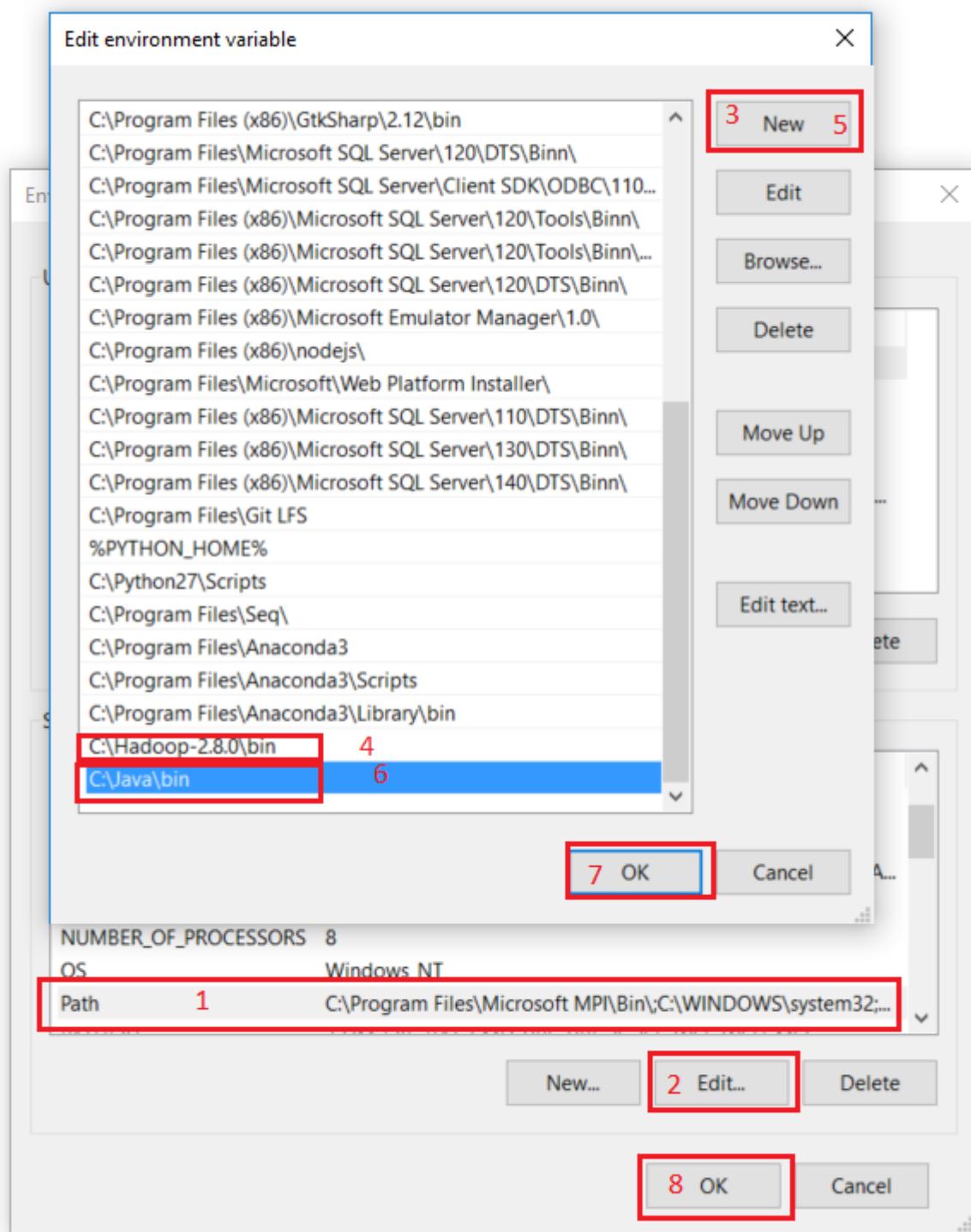
4. Set the path HADOOP_HOME Environment variable on windows 10(see Step 1,2,3 and 4 below).



5. Set the path JAVA_HOME Environment variable on windows 10(see Step 1,2,3 and 4 below).



6. Next we set the Hadoop bin directory path and JAVA bin directory



path.

Configuration

1. Edit file **C:/Hadoop-2.8.0/etc/hadoop/core-site.xml**, paste below xml paragraph and save this file.

```
<configuration>
  <property>
    <name>fs.defaultFS</name>
    <value>hdfs://localhost:9000</value>
  </property>
</configuration>
```

2. Rename "mapred-site.xml.template" to "mapred-site.xml" and edit this file **C:/Hadoop-2.8.0/etc/hadoop/mapred-site.xml**, paste below xml paragraph and save this file.

```
<configuration>
  <property>
    <name>mapreduce.framework.name</name>
    <value>yarn</value>
  </property>
</configuration>
```

3. Create folder "**data**" under "**C:\Hadoop-2.8.0**"

- Create folder "**datanode**" under "**C:\Hadoop-2.8.0\data**"
- Create folder "**namenode**" under "**C:\Hadoop-2.8.0\data**"

☐ Name	Date modified	Type	Size
bin	7/20/2017 2:14 PM	File folder	
☒ data	7/20/2017 2:47 PM	File folder	
etc	7/20/2017 2:14 PM	File folder	
include	7/20/2017 2:14 PM	File folder	
lib	7/20/2017 2:14 PM	File folder	
libexec	7/20/2017 2:14 PM	File folder	
sbin	7/20/2017 2:14 PM	File folder	
share	7/20/2017 2:20 PM	File folder	
LICENSE.txt	3/17/2017 10:31 AM	TXT File	97 KB
NOTICE.txt	3/17/2017 10:31 AM	TXT File	16 KB
README.txt	3/17/2017 10:31 AM	TXT File	2 KB

4. Edit file **C:\Hadoop-2.8.0/etc/hadoop/hdfs-site.xml**, paste below xml paragraph and save this file.

```
<configuration>
  <property>
    <name>dfs.replication</name>
    <value>1</value>
  </property>
  <property>
    <name>dfs.namenode.name.dir</name>
    <value>C:\hadoop-2.8.0\data\namenode</value>
  </property>
  <property>
    <name>dfs.datanode.data.dir</name>
    <value>C:\hadoop-2.8.0\data\datanode</value>
  </property>
</configuration>
```

5. Edit file **C:/Hadoop-2.8.0/etc/hadoop/yarn-site.xml**, paste below xml paragraph and save this file.

```
<configuration>
  <property>
    <name>yarn.nodemanager.aux-services</name>
    <value>mapreduce_shuffle</value>
  </property>
```

```
<property>
  <name>yarn.nodemanager.auxservices.mapreduce.shuffle.class</name>
  <value>org.apache.hadoop.mapred.ShuffleHandler</value>
</property>
</configuration>
```

6. Edit file **C:/Hadoop-2.8.0/etc/hadoop/hadoop-env.cmd** by closing the command line "**JAVA_HOME=%JAVA_HOME%**" instead of set "**JAVA_HOME=C:\Java**" (On C:\java this is path to file jdk.18.0)

```
@rem The java implementation to use.  Required.
@rem set JAVA_HOME=%JAVA_HOME%
set JAVA_HOME=C:\java
```

Hadoop Configuration

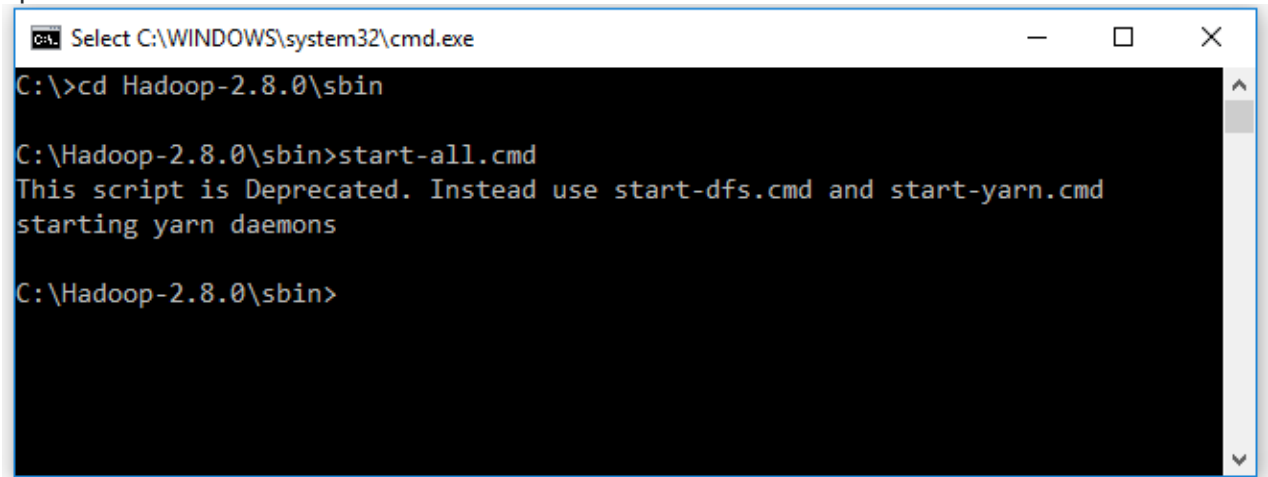
1. Dowload file Hadoop Configuration.zip
(Link: <https://github.com/MuhammadBilalYar/HADOOP-INSTALLATION-ON-WINDOW-10/blob/master/Hadoop%20Configuration.zip>)
2. Delete file bin on C:\Hadoop-2.8.0\bin, replaced by file bin on file just download (from Hadoop Configuration.zip).
3. Open cmd and typing command "**hdfs namenode -format**". You will see

```
C:\WINDOWS\system32\cmd.exe
17/07/20 15:38:21 INFO util.GSet: VM type = 64-bit
17/07/20 15:38:21 INFO util.GSet: 0.25% max memory 889 MB = 2.2 MB
17/07/20 15:38:21 INFO util.GSet: capacity = 2^18 = 262144 entries
17/07/20 15:38:21 INFO namenode.FSNamesystem: dfs.namenode.safemode.threshold-pct = 0.9990000128746033
17/07/20 15:38:21 INFO namenode.FSNamesystem: dfs.namenode.safemode.min.datanodes = 0
17/07/20 15:38:21 INFO namenode.FSNamesystem: dfs.namenode.safemode.extension = 30000
17/07/20 15:38:21 INFO metrics.TopMetrics: NNTop conf: dfs.namenode.top.window.num.buckets = 10
17/07/20 15:38:21 INFO metrics.TopMetrics: NNTop conf: dfs.namenode.top.num.users = 10
17/07/20 15:38:21 INFO metrics.TopMetrics: NNTop conf: dfs.namenode.top.windows.minutes = 1,5,25
17/07/20 15:38:21 INFO namenode.FSNamesystem: Retry cache on namenode is enabled
17/07/20 15:38:21 INFO namenode.FSNamesystem: Retry cache will use 0.03 of total heap and retry cache entry expiry time is 600000 millis
17/07/20 15:38:21 INFO util.GSet: Computing capacity for map NameNodeRetryCache
17/07/20 15:38:21 INFO util.GSet: VM type = 64-bit
17/07/20 15:38:21 INFO util.GSet: 0.029999999329447746% max memory 889 MB = 273.1 KB
17/07/20 15:38:21 INFO util.GSet: capacity = 2^15 = 32768 entries
17/07/20 15:38:26 INFO namenode.FSImage: Allocated new BlockPoolId: BP-1080504939-192.168.68.1-1500547106632
17/07/20 15:38:26 INFO common.Storage: Storage directory C:\Hadoop-2.8.0\data\namenode has been successfully formatted.
17/07/20 15:38:26 INFO namenode.FSImageFormatProtobuf: Saving image file C:\Hadoop-2.8.0\data\namenode\current\fsimage.ckpt_000000000000000000 using no compression
17/07/20 15:38:26 INFO namenode.FSImageFormatProtobuf: Image file C:\Hadoop-2.8.0\data\namenode\current\fsimage.ckpt_000000000000000000 of size 330 bytes saved in 0 seconds.
17/07/20 15:38:26 INFO namenode.NNStorageRetentionManager: Going to retain 1 images with txid >= 0
17/07/20 15:38:26 INFO util.ExitUtil: Exiting with status 0
17/07/20 15:38:26 INFO namenode.NameNode: SHUTDOWN_MSG:
/*****
SHUTDOWN_MSG: Shutting down NameNode at [REDACTED]
*****/
C:\Users\Muhammad.bilal>
```

Testing

1. Open cmd and change directory to "C:\Hadoop-2.8.0\sbin" and type "**start-all.cmd**" to start

apache.



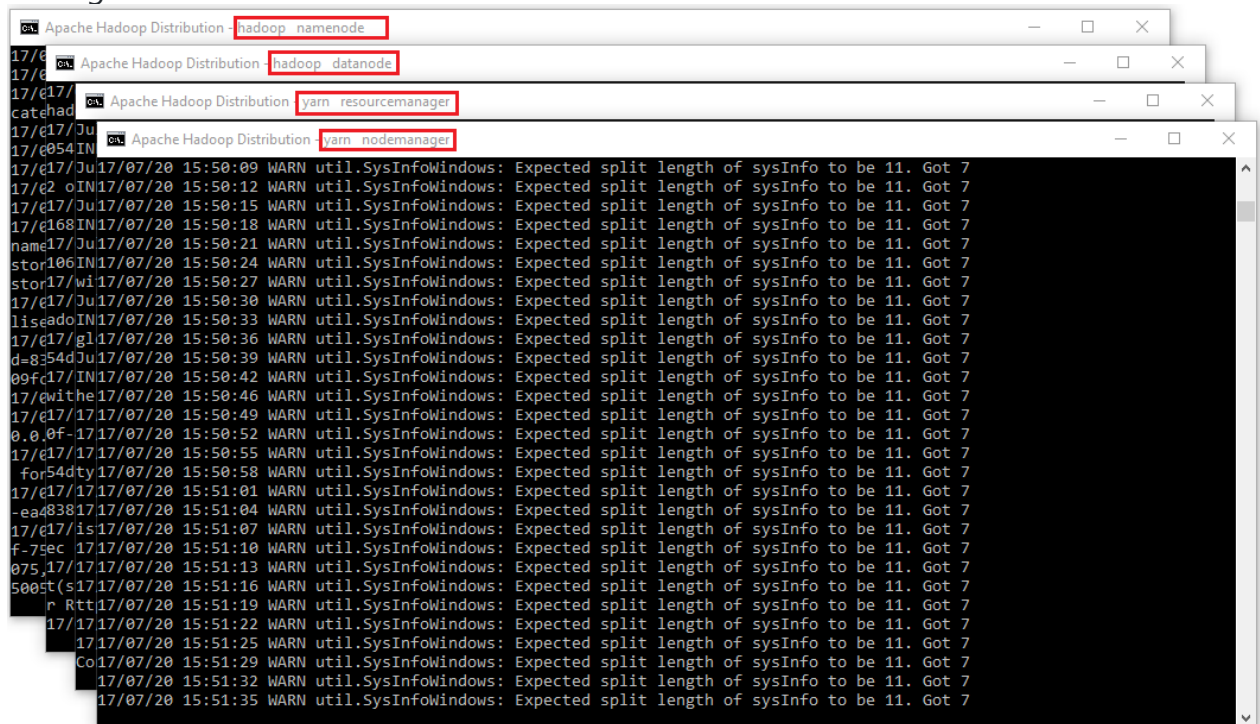
```
C:\>cd Hadoop-2.8.0\sbin

C:\Hadoop-2.8.0\sbin>start-all.cmd
This script is Deprecated. Instead use start-dfs.cmd and start-yarn.cmd
starting yarn daemons

C:\Hadoop-2.8.0\sbin>
```

2. Make sure these apps are running

- Hadoop Namenode
- Hadoop datanode
- YARN Resourc Manager
- YARN Node Manager



```
17/07/20 15:50:09 WARN util.SysInfoWindows: Expected split length of sysInfo to be 11. Got 7
17/07/20 15:50:12 WARN util.SysInfoWindows: Expected split length of sysInfo to be 11. Got 7
17/07/20 15:50:15 WARN util.SysInfoWindows: Expected split length of sysInfo to be 11. Got 7
17/07/20 15:50:18 WARN util.SysInfoWindows: Expected split length of sysInfo to be 11. Got 7
17/07/20 15:50:21 WARN util.SysInfoWindows: Expected split length of sysInfo to be 11. Got 7
17/07/20 15:50:24 WARN util.SysInfoWindows: Expected split length of sysInfo to be 11. Got 7
17/07/20 15:50:27 WARN util.SysInfoWindows: Expected split length of sysInfo to be 11. Got 7
17/07/20 15:50:30 WARN util.SysInfoWindows: Expected split length of sysInfo to be 11. Got 7
17/07/20 15:50:33 WARN util.SysInfoWindows: Expected split length of sysInfo to be 11. Got 7
17/07/20 15:50:36 WARN util.SysInfoWindows: Expected split length of sysInfo to be 11. Got 7
17/07/20 15:50:39 WARN util.SysInfoWindows: Expected split length of sysInfo to be 11. Got 7
17/07/20 15:50:42 WARN util.SysInfoWindows: Expected split length of sysInfo to be 11. Got 7
17/07/20 15:50:46 WARN util.SysInfoWindows: Expected split length of sysInfo to be 11. Got 7
17/07/20 15:50:49 WARN util.SysInfoWindows: Expected split length of sysInfo to be 11. Got 7
17/07/20 15:50:52 WARN util.SysInfoWindows: Expected split length of sysInfo to be 11. Got 7
17/07/20 15:50:55 WARN util.SysInfoWindows: Expected split length of sysInfo to be 11. Got 7
17/07/20 15:50:58 WARN util.SysInfoWindows: Expected split length of sysInfo to be 11. Got 7
17/07/20 15:51:01 WARN util.SysInfoWindows: Expected split length of sysInfo to be 11. Got 7
17/07/20 15:51:04 WARN util.SysInfoWindows: Expected split length of sysInfo to be 11. Got 7
17/07/20 15:51:07 WARN util.SysInfoWindows: Expected split length of sysInfo to be 11. Got 7
17/07/20 15:51:10 WARN util.SysInfoWindows: Expected split length of sysInfo to be 11. Got 7
17/07/20 15:51:13 WARN util.SysInfoWindows: Expected split length of sysInfo to be 11. Got 7
17/07/20 15:51:16 WARN util.SysInfoWindows: Expected split length of sysInfo to be 11. Got 7
17/07/20 15:51:19 WARN util.SysInfoWindows: Expected split length of sysInfo to be 11. Got 7
17/07/20 15:51:22 WARN util.SysInfoWindows: Expected split length of sysInfo to be 11. Got 7
17/07/20 15:51:25 WARN util.SysInfoWindows: Expected split length of sysInfo to be 11. Got 7
17/07/20 15:51:29 WARN util.SysInfoWindows: Expected split length of sysInfo to be 11. Got 7
17/07/20 15:51:32 WARN util.SysInfoWindows: Expected split length of sysInfo to be 11. Got 7
17/07/20 15:51:35 WARN util.SysInfoWindows: Expected split length of sysInfo to be 11. Got 7
```


3. Open: <http://localhost:8088>

Cluster

About

Nodes

Node Labels

Applications

NEW

NEW SAVING

SUBMITTED

ACCEPTED

RUNNING

FINISHED

FAILED

KILLED

Scheduler

Tools

All Applications

Logged in as: dr.who

Cluster Metrics

Apps Submitted	Apps Pending	Apps Running	Apps Completed	Containers Running	Memory Used	Memory Total	Memory Reserved	VCores Used	VCores Total	VCores Reserved
0	0	0	0	0	0 B	8 GB	0 B	0	8	0

Cluster Nodes Metrics

Active Nodes	Decommissioning Nodes	Decommissioned Nodes	Lost Nodes	Unhealthy Nodes	Rebooted Nodes	Shutdown Nodes
1	0	0	0	0	0	0

Scheduler Metrics

Scheduler Type	Scheduling Resource Type	Minimum Allocation	Maximum Allocation	Maximum Cluster Application Priority
Capacity Scheduler	[MEMORY]	<memory:1024, vCores:1>	<memory:8192, vCores:4>	0

Show 20 entries

Search:

ID	User	Name	Application Type	Queue	Application Priority	StartTime	FinishTime	State	FinalStatus	Running Containers	Allocated CPU VCoers	Allocated Memory MB	% of Queue	% of Cluster	Progress	Tracking UI	Blacklisted Nodes
No data available in table																	

Showing 0 to 0 of 0 entries

First Previous Next Last